THE LIMIT TOPOLOGY ON THE COHOMOLOGY OF AN INVERSE SYSTEM OF SHEAVES IS I-ADIC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author marks one structural verification as omitted: at source line 4905 the sentence recording that the direct sum of the E_n is a graded submodule and that the map to the direct sum of the F^n/F^{n+1} is graded ends with 'details omitted'. That verification concerns the bookkeeping of the graded structures, and the load-bearing steps of the argument, the finite generation, the identity IF^n = F^{n+1} and the open mapping lemma, are all written out inside the counted span. The omission is recorded here rather than repaired.

Editorial note. The counted statement carries the author's footnote on the uniform form of the conclusion; it lies inside the statement span and is delivered as written.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology for a topological space. That statement is the special case of this one, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The single labelled reference occurring inside the counted spans points to another chapter of the Project; it is carried in assembly.external_references as a non-load-bearing navigation pointer recorded at the invoking line.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- I-adic and limit topologies on the limit of an inverse system of modules. Same chapter, the comparison of the I-adic topology and the limit topology on M = lim H^p(C, F_n). Supplies both topologies whose equality is asserted, and the chain F^n = ker(M -> H^p(C, F_n)) of neighbourhoods of the limit topology.
- Graded module structure over the ring of the powers of I. Same chapter, the graded module structure of the direct sum of the H^p(C, I^n F_{n+1}) over the direct sum of the I^n/I^{n+1}. Supplies the ascending chain condition hypothesis and the graded generators used in the proof.
- Open mapping lemma for complete linearly topologized modules. More on Algebra, the open mapping lemma applied to u : (F^n)^{direct sum r} -> F^{n+1}. Supplies the step that upgrades density of the image to surjectivity of u.

Source reference. Original lemma 0CQW: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4935–4935. Cross-chapter navigation pointer invoked at proof lines 4934-4935, where the author applies the open mapping lemma

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for complete linearly topologized modules to conclude that the map u on the dense image is open. The target chapter is not part of this card's source tree, so the pointer records the invoking line; it only resolves the reference for the reader, and no bibliography citation is involved.

Lemma 0.1. *Let I be an ideal of a ring A. Let $\mathcal{C}$ be a site. Let*

$$\ldots \to \mathcal{F}_3 \to \mathcal{F}_2 \to \mathcal{F}_1$$

be an inverse system of sheaves of A-modules on $\mathcal{C}$ such that $\mathcal{F}_n = \mathcal{F}_{n+1}/I^n\mathcal{F}_{n+1}$. Let $p \geq 0$. Assume

$$\bigoplus\nolimits_{n\geq 0} H^p(\mathcal{C}, I^n\mathcal{F}_{n+1})$$

satisfies the ascending chain condition as a graded $\bigoplus_{n\geq 0} I^n/I^{n+1}$-module. Then the limit topology on $M = \lim H^p(\mathcal{C}, \mathcal{F}_n)$ is the I-adic topology.

Proof. Set $F^n = \mathrm{Ker}(M \to H^p(\mathcal{C}, \mathcal{F}_n))$ for $n \geq 1$ and $F^0 = M$. Observe that $IF^n \subset F^{n+1}$. In particular $I^nM \subset F^n$. Hence the I-adic topology is finer than the limit topology. For the converse, we will show that given n there exists an $m \geq n$ such that $F^m \subset I^nM$[1]. We have injective maps

$$F^n/F^{n+1} \longrightarrow H^p(\mathcal{C}, \mathcal{F}_{n+1})$$

whose image is contained in the image of $H^p(\mathcal{C}, I^n\mathcal{F}_{n+1}) \to H^p(\mathcal{C}, \mathcal{F}_{n+1})$. Denote

$$E_n \subset H^p(\mathcal{C}, I^n\mathcal{F}_{n+1})$$

the inverse image of F^n/F^{n+1}. Then $\bigoplus E_n$ is a graded $\bigoplus I^n/I^{n+1}$-submodule of $\bigoplus H^p(\mathcal{C}, I^n\mathcal{F}_{n+1})$ and $\bigoplus E_n \to \bigoplus F^n/F^{n+1}$ is a homomorphism of graded modules; details omitted. By assumption $\bigoplus E_n$ is generated by finitely many homogeneous elements over $\bigoplus I^n/I^{n+1}$. Since $E_n \to F^n/F^{n+1}$ is surjective, we see that the same thing is true of $\bigoplus F^n/F^{n+1}$. Hence we can find r and $c_1, \ldots, c_r \geq 0$ and $a_i \in F^{c_i}$ whose images in $\bigoplus F^n/F^{n+1}$ generate. Set $c = \max(c_i)$.

For $n \geq c$ we claim that $IF^n = F^{n+1}$. The claim shows that $F^{n+c} = I^nF^c \subset I^nM$ as desired. To prove the claim suppose $a \in F^{n+1}$. The image of a in F^{n+1}/F^{n+2} is a linear combination of our a_i. Therefore $a - \sum f_ia_i \in F^{n+2}$ for some $f_i \in I^{n+1-c_i}$. Since $I^{n+1-c_i} = I \cdot I^{n-c_i}$ as $n \geq c_i$ we can write $f_i = \sum g_{i,j}h_{i,j}$ with $g_{i,j} \in I$ and $h_{i,j}a_i \in F^n$. Thus we see that $F^{n+1} = F^{n+2} + IF^n$. A simple induction argument gives $F^{n+1} = F^{n+e} + IF^n$ for all $e > 0$. It follows that IF^n is dense in F^{n+1}. Choose generators $k_1, \ldots, k_r$ of I and consider the continuous map

$$u : (F^n)^{\oplus r} \longrightarrow F^{n+1}, \quad (x_1, \ldots, x_r) \mapsto \sum k_ix_i$$

(in the limit topology). By the above the image of $(F^m)^{\oplus r}$ under u is dense in F^{m+1} for all $m \geq n$. By the open mapping lemma (More on Algebra, Lemma 0CQW) we find that u is open. Hence u is surjective. Hence $IF^n = F^{n+1}$ for $n \geq c$. This concludes the proof. □

[1] In fact, there exist a $c \geq 0$ such that $F^{n+c} \subset I^nM$ for all n.